EXPERIMENTAL - bloxberg Blockchain

Certification of metadata file on the bloxberg blockchain

Internal JSON | External JSON

Dataset: FRIM - Fruit Integrative Modelling

bloxberg Server URL:
https://certify.bloxberg.org/

Wallet bloxberg:
0xC604ffa8adE14dc9A22B6B19bdFC07E489156E53

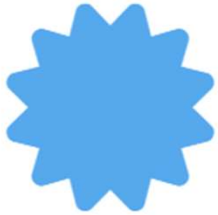
API bloxberg:

Publish

bloxberg

Certification

Certify your Research Data on the blockchain.



This protects your intellectual property and safeguards against potential future scooping by generating a unique identifier from your data that is then stored on the chain. A certificate is then created to provide proof that this information belongs to you or your organization.

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API bloxberg:
[redacted]

Publish

1 Create the request

- Metadata file — META_frim1.json
- bloxberg metadata — bloxberg_metadata_frim1.json
- bloxberg query — bloxberg_query_frim1.json

Metadata File


```
{
  "profile": "https://pmb-bordeaux.fr/maggot/conf/maggot-schema.json",
  "title": "frim1",
  "fulltitle": "FRIM - Fruit Integrative Modelling",
  "license": "Etalab V2.0 license",
  "depositor": "Jacob Daniel",
  "publication_idnumber": "10.1104/pp.113.231241",
  "publication_url": "https://doi.org/10.1104/pp.113.231241",
  "description": "The project aimed to build a virtual tomato fruit that enables the prediction of metabolite levels given genetic and environmental inputs, by an iterative process between laboratories which combine expertise in fruit biology, ecophysiology, theoretical and experimental biochemistry, and biotechnology.",
  "publication_citation": "Biais B, Bénard C, Beauvoit B, Colombié S, Prodhomme D, Ménard G, Bernillon S, Gehl B, Gautier H, Ballias P, Mazat J-P, Sweetlove L, Génard M, Gibon Y. 2014. Remarkable reproducibility of enzyme activity profiles in tomato fruits grown under contrasting environments provides a roadmap for studies of fruit metabolism. Plant Physiology 164, 1204-1221",
  "status": "Processed",
  "access_rights": "Public",
  "publication_idtype": "doi",
  "contacts": [
    "Gibon Yves"
  ],
  "authors": [
    "Bénard Camille",
    "Biais Benoit",
    "Beauvoit Bertrand",
    "Colombié Sophie"
  ],
  "collectors": [
    "Bénard Camille",
    "Biais Benoit",
    "Ballias Patricia",
    "Maucourt Mickaël"
  ],
  "curators": [
    "Moing Annick",
    "Jacob Daniel"
  ],
  "leader": [
    "Gibon Yves"
  ],
  "wpleader": [
    "Vercambre Gilles"
  ],
  "producer": [
    "Bordeaux Metabolome"
```

Metadata of the Metadata File

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API bloxberg:
[redacted]

Publish

1 Create the request

- Metadata file — META_frim1.json
- bloxberg metadata — bloxberg_metadata_frim1.json
- bloxberg query — bloxberg_query_frim1.json

Same as


```
{
  "@context": {
    "dc": "http://purl.org/dc/terms/"
  },
  "@type": "Dataset",
  "dc:creator": "https://orcid.org/0000-0001-8161-1089",
  "dc:language": "en",
  "dc:format": "application/json",
  "dc:title": "FRIM - Fruit Integrative Modelling",
  "dc:subject": "https://id.loc.gov/authorities/subjects/sh96000740",
  "dc:date": "2023-11-27T16:17:11.3629640Z",
  "dc:description": "The project aimed to build a virtual tomato fruit that enables
  the prediction of metabolite levels given genetic and environmental inputs, by an
  iterative process between laboratories which combine expertise in fruit biology,
  ecophysiology, theoretical and experimental biochemistry, and biotechnology.",
  "dc:publisher": [
    "https://ror.org/00ccpdp17"
  ],
  "dc:rights": "http://spdx.org/licenses/CC-BY-NC-SA-4.0.json",
  "dc:identifier": {
    "@value": "https://pmb-bordeaux.fr/maggot/metadata/frim1?format=maggot",
    "@type": "http://purl.org/dc/terms/URI"
  }
}
```

EXPERIMENTAL - bloxberg Blockchain

Certification of metadata file on the bloxberg blockchain

Internal JSON | External JSON

Dataset: FRIM - Fruit Integrative Modelling

bloxberg Server URL:
https://certify.bloxberg.org/

Wallet bloxberg:
0xC604ffa8adE14dc9A22B6B19bdFC07E489156E53

API bloxberg:
[redacted]

Publish

1 Create the request

- Metadata file — META_frim1.json
- bloxberg metadata — bloxberg_metadata_frim1.json
- bloxberg query — bloxberg_query_frim1.json

Same as

Hash sha256

Metadata of the Metadata File


```
{
  "@context": {
    "dc": "http://purl.org/dc/terms/"
  },
  "@type": "Dataset",
  "dc:creator": "https://orcid.org/0000-0001-8161-1089",
  "dc:language": "en",
  "dc:format": "application/json",
  "dc:title": "FRIM - Fruit Integrative Modelling",
  "dc:subject": "https://id.loc.gov/authorities/subjects/sh96000740",
  "dc:date": "2023-11-27T16:17:11.3629640Z",
  "dc:description": "The project aimed to build a virtual tomato fruit that enables
  the prediction of metabolite levels given genetic and environmental inputs, by an
  iterative process between laboratories which combine expertise in fruit biology,
  ecophysiology, theoretical and experimental biochemistry, and biotechnology.",
  "dc:publisher": [
    "https://ror.org/00ccpdp17"
  ],
  "dc:rights": "http://spdx.org/licenses/CC-BY-NC-SA-4.0.json",
  "dc:identifier": {
    "@value": "https://pmb-bordeaux.fr/maggot/metadata/frim1?format=maggot",
    "@type": "http://purl.org/dc/terms/URI"
  }
}
```


Embedded

Request


```
{
  "publicKey": "0xC604ffa8adE14dc9A22B6B19bdFC07E489156E53",
  "crid": [
    "ae008cc8ebd20834befa9442db0e59d53effe5ba6cbb8612f9544f8be2fec2b0"
  ],
  "cridType": "sha2-256",
  "enableIPFS": false,
  "metadataJson": "{ \"@context\": { \"dc\": \"http://purl.org/dc/terms/\" }, \"@type\": \"Dataset\",
  \"dc:creator\": \"https://orcid.org/0000-0001-8161-1089\", \"dc:language\": \"en\", \"dc:format\"
  \"application/json\", \"dc:title\": \"FRIM - Fruit Integrative Modelling\", \"dc:subject\":
  \"https://id.loc.gov/authorities/subjects/sh96000740\", \"dc:date\": \"2023-11-
  27T16:17:11.3629640Z\", \"dc:description\": \"The project aimed to build a virtual tomato fruit
  that enables the prediction of metabolite levels given genetic and environmental inputs, by an
  iterative process between laboratories which combine expertise in fruit biology, ecophysiology,
  theoretical and experimental biochemistry, and biotechnology.\", \"dc:publisher\":
  [\"https://ror.org/00ccpdp17\"], \"dc:rights\":\"http://spdx.org/licenses/CC-BY-NC-SA-4.0.json\",
  \"dc:identifier\": { \"@value\": \"https://pmb-bordeaux.fr/maggot/metadata/frim1?format=maggot\",
  \"@type\": \"http://purl.org/dc/terms/URI\" }}"
}
```

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Wallet bloxberg:
0xC604ffa8adE14dc9A22B6B19bdFC07E489156E53

API bloxberg:
[redacted]

Publish

2 Submit the request

- Metadata file — META_frim1.json
- bloxberg metadata — bloxberg_metadata_frim1.json
- bloxberg query — bloxberg_query_frim1.json
- bloxberg response — bloxberg_response_frim1.json


```
[
  {
    "@context": [
      "https://www.w3.org/2018/credentials/v1",
      "https://w3id.org/bloxberg/schema/research_object_certificate_v1"
    ],
    "type": [
      "VerifiableCredential",
      "BloxbergCredential"
    ],
    ...
    "proof": {
      "type": "MerkleProof2019",
      "created": "2023-10-31T08:18:52.993006",
      "proofValue": "z7veGu1qoKR3AS5A6VJKrNwcowzhzPvSz9MSMz8661R7iJ1FtnmVUbSJLwna4b9f4MNsJ634P7m51PW
      KYmUhEowJFpyuYDBmx3AT8bFYP3cQ22kDea7gTYiLRdPtVApgUDwDg86MWozSE4o76FhfDWKUKnTetca5f63Gebifrk8S9
      Ro1B4yN4QxF7PRqTPESaiUadffofXiwbxtTiQ7ipTQPuk5auhJbt1Sz4wCj3ELAfL4pYozYBUPVrNJgFWiVQxeTHR1NBqh
      C9JHxmrJfkuXJvL15aJEHWW2XhE86zYuKHuSPmUrCm8",
      "proofPurpose": "assertionMethod",
      "verificationMethod": "ecdsa-koblitz-pubkey:0xD748BF41264b90609346092316964 3f45BDbC32e",
      "ens_name": "mpdl.berg"
    }
  }
]
```

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Dataset: FRIM - Fruit Integrative Modelling

bloxberg Server URL:
https://certify.bloxberg.org/

Wallet bloxberg:
0xC604ffa8adE14dc9A22B6B19bdFC07E489156E53

API bloxberg:
[redacted]

Publish

2 Submit the request

- Metadata file — META_frim1.json
- bloxberg metadata — bloxberg_metadata_frim1.json
- bloxberg query — bloxberg_query_frim1.json
- bloxberg response — bloxberg_response_frim1.json


```
[
  {
    "@context": [
      "https://www.w3.org/2018/credentials/v1",
      "https://w3id.org/bloxberg/schema/research_object_certificate_v1"
    ],
    "type": [
      "VerifiableCredential",
      "BloxbergCredential"
    ],
    ...
    "proof": {
      "type": "MerkleProof2019",
      "created": "2023-10-31T08:18:52.993006",
      "proofValue": "z7veGu1qoKR3AS5A6VJKrNwcowzhzPvSz9MSMz8661R7iJ1FtnmVUbSJLwna4b9f4MNsJ634P7m51PW
      KYmUhEowJFpyuYDBmx3AT8bFYP3cQ22kDea7gTYiLRdPtVApgUDwDg86MWozSE4o76FhfDWKUKnTetca5f63Gebifrk8S9
      Ro1B4yN4QxF7PRqTPESaiUadffofXiwbxtTiQ7ipTQPuk5auhJbt1Sz4wCj3ELAfL4pYozYBUPVrNJgFWiVQxeTHR1NBqh
      C9JHxmrJfkuXJvL15aJEHWW2XhE86zYuKHuSPmUrCm8",
      "proofPurpose": "assertionMethod",
      "verificationMethod": "ecdsa-koblitz-pubkey:0xD748BF41264b906093460923169643f45BDbC32e",
      "ens_name": "mpdl.berg"
    }
  }
]
```


Proof of Authority

Transactions and blocks are validated by approved accounts, known as validators.

However the process is automated and does not require validators to be constantly monitoring their computers.

https://en.wikipedia.org/wiki/Proof_of_authority

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Wallet bloxberg:
0xC604ffa8adE14dc9A22B6B19bdFC07E489156E53

API bloxberg:
[redacted]

Publish

3 Retrieve the certificate

- Metadata file — META_frim1.json
- bloxberg metadata — bloxberg_metadata_frim1.json
- bloxberg query — bloxberg_query_frim1.json
- bloxberg response — bloxberg_response_frim1.json
- bloxberg certificate — bloxberg_certificate_frim1.zip

Success : the metadata file has been certified on the bloxberg blockchain
Download the ZIP which contains the corresponding certificate as a PDF file


```
[
  {
    "@context": [
      "https://www.w3.org/2018/credentials/v1",
      "https://w3id.org/bloxberg/schema/research_object_certificate_v1"
    ],
    "type": [
      "VerifiableCredential",
      "BloxbergCredential"
    ],
    ...
    "proof": {
      "type": "MerkleProof2019",
      "created": "2023-10-31T08:18:52.993006",
      "proofValue": "z7veGu1qoKR3AS5A6VJKrNwcowzhzPvSz9MSMz8661R7iJ1FtnmVUbSJLwna4b9f4MNsJ634P7m51PW
      KYmUhEowJFpyuYDBmx3AT8bFYP3cQ22kDea7gTYiLRdPtVApgUDwDg86MWozSE4o76FhfDWKUKnTetca5f63Gebifrk8S9
      Ro1B4yN4QxF7PRqTPESaiUadffofXiwbxtTiQ7ipTQPuk5auhJbt1Sz4wCj3ELAfL4pYozYBUPVrNJgFWiVQxeTHR1NBqh
      C9JHxmrJfkuXJvL15aJEHWW2XhE86zYuKHuSPmUrCm8",
      "proofPurpose": "assertionMethod",
      "verificationMethod": "ecdsa-koblitz-pubkey:0xD748BF41264b90609346092316964 3f45BDbC32e",
      "ens_name": "mpdl.berg"
    }
  }
]
```